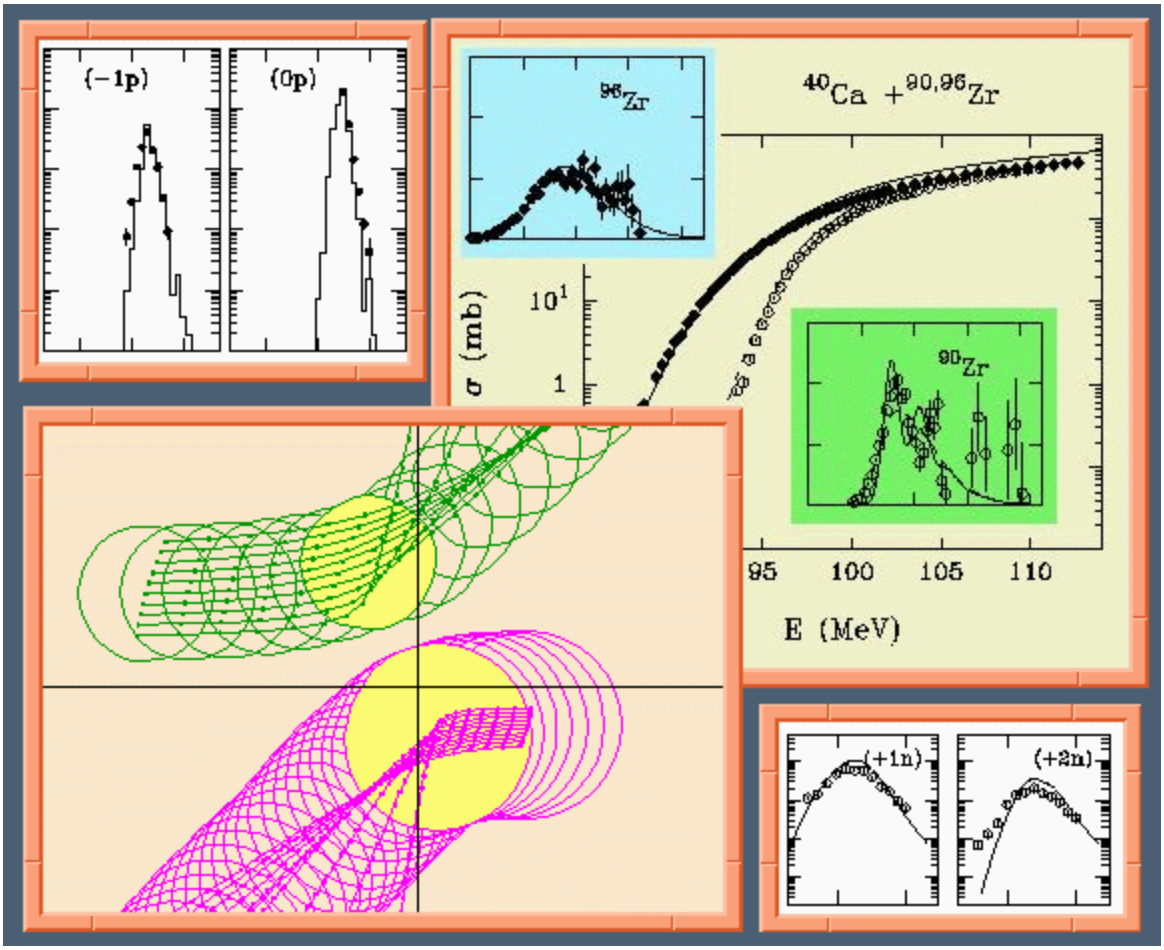
GRAZING_9
A Fortran program for estimating reactions in collision between Heavy Nuclei

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Abstract:

GRAZING_9 is an interactive Fortran program that evaluates the reactions to be expected in a collision between two heavy nuclei at moderate bombarding energies. It calculates the cross section for the distribution in mass and charge, kinetic energy and excitation energy of the two scattered nuclei. It also calculates the cross section leading to capture including coupled channel effect in the tunneling of the fusion barrier. It uses the NAG library Mark 18.

To download the EXE

Here you find the zipped tar file that contains the executable of the GRAZING program that should run on any LINUX (there is a version at 32bits and one at 64bits).
To install unzip and untar the file in any location of your choice, there you find the grazing_9 (the exe files at 32bits and the one at 64bits) and a directory data containing all the file needed by GRAZING.

To run the program you MUST first define the enviromental variable GRAZING_DIR that refer to the data directory. In the bash shell:

```
(Place these lines in .bash_profile)

GRAZING_DIR='my_grazing_dir/data'
export GRAZING_DIR
```

Let me know if it does not wotk

To download FULL

Here you find the zipped tar file that contains the program: grazing_9 (393 Kb - June 2002). Notice this is the last version with quite few upgrades done by myself so the mistakes rest on me and not on the author. If you have problems please do not esitate to contact me (see at the bottom).

Instruction to install the program.

This distribution of **GRAZING_9** has been tested to run both on **LINUX** (redhat 6.2) and on **ALPHA** (Compaq TRU64-UNIX). The program uses the **NAG library (Mark 18).**

Copy the .gz file in the directory you want to place grazing_9. Then issue the following commands to unzip and untar the file:

```
>gunzip grazing.tar.gz
>tar xvf grazing.tar
```

After these operations you have a new directory grazing that contains all the relevant files for grazing_9 and three (3) subdirectories:

grazing/doc	contains some documentation about the program
grazing/pg	contains the plotting program gzp2. You may find it usefull, otherwise just discart it.
grazing/examples	contains few scripts for running grazing_9. There is also e file .inp for plotting with gzp2.

Now follows these steps to satisfy the local setting and the architecture you are interested in:

LINUX - ALPHA

1 - **grazing_file.icl**
modify the file according to you local setting (use the full path name). The files referred contain informations about exces masses of the nuclei and the energy and strenght of the low lying states

2 - **makefile**
modify the file to satisfy your local setting for the NAG library and other global definitions like compiler and the use of other routines. Notise that these corrections appear at the beginning of the file and at the end.

3 - issue the command
>make
If everithing is correct you should have the file grazing_9 that contains the executable.

4 - Make sure you have installed on your machine the graphical PGPLOT
library. Go to the subdirectory pg. Modify the script flpg to satisfy
yout local setting and then issue the command:
>flpg gzp2
You should have the plotting program gzp2

5 - Go to the subdirectory examples, modify the file **bd_com**
and **fus_com** to satisfy your local setting

6 - Execute both scripts. You should have two new files .dat containing the
fusion cross sections and the barrier distribution

7 - issue the following command
>../pg/gzp2
pg_fus.inp
/XWINDOW
a new window should appear on the screen with the
excitation function and the barrier distribution you just calculated.
To close the window pres a **C_RET** from the command screen.
If you feel satisfied with gzp2 you may learn more about it consulting
the page: **http:/www.to.infn.it/~nanni/gzp2.html**.

For any problems please send mail to nanni@to.infn.it
